

2024



HKBGE Enhances Upcoming Crypto Exchange Platform with Generative AI Chatbot on AWS

Hong Kong BGE Limited (HKBGE) is developing a generative AI customer service chatbot on Amazon EC2 G5 instances, aiming to improve scalability, personalize responses, and redirect resources to staff development.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

70%

cost savings redirected to staff training

I



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Improved customer onboarding

with personalized chatbot responses



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Greater productivity



Overview

[HKBGE](#) is establishing a compliant and trusted virtual asset trading platform for clients to transact and trade virtual assets together with the company's secure custody services. To personalize responses to customers' questions, HKBGE collaborated with AWS to create a generative AI chatbot based on Meta's Llama 2 large language model.

The chatbot, powered by [Amazon EC2](#) G5 instances with NVIDIA A10G Tensor Core GPUs, is expected to deliver a better customer onboarding experience, providing thousands of personalized responses daily. Furthermore, HKBGE anticipates reducing support costs by 70 percent, freeing up resources to invest in staff training and tailored customer service.



Opportunity | Providing Crypto Traders Personalized Chatbot Support

[HKE Holdings](#), a listed fintech group focused on the Asia-Pacific region, established [Hong Kong BGE](#) Limited (HKBGE) to offer digital asset trading solutions for both retail and professional investors. In 2021, HKBGE applied for a virtual asset trading platform license under Hong Kong's Securities and Futures Commission (SFC) to offer crypto exchange services.

Upon the anticipated launch of its trading platform, HKBGE foresees receiving a large volume of thousands of interested customers once the license is granted by the SFC. The most common inquiries relate to routine tasks such as account opening standards, checking balances, and deposit/withdrawals.



Initial inquiries are expected to peak in the first few weeks post-launch. As the business expands, HKBGE anticipates a proportional increase in queries over time.

To help deliver a better customer experience, HKBGE looked to develop a generative artificial intelligence (AI) [chatbot](#) to provide personalized responses to frequently asked questions (FAQ) as customers visit the platform. The solution is designed to minimize resolution times and free up customer support staff to focus on more complex tasks and critical customer interactions.

Solution | Maximizing Chatbot Performance and Platform Security

Since the license application in 2021, HKBGE has collaborated extensively with Amazon Web Services (AWS) on the development of its trading platform. Later in 2023, HKBGE's IT team worked with AWS Solutions Architects to develop a retrieval-augmented generation (RAG) chatbot, which combines the advantages of retrieval-based models with generative AI models to provide more contextually relevant responses.

As part of the development process, the CS & IT team worked with AWS on the RAG chatbot's search capability. This included looking at different databases, such as the open-source VectorDB. The two parties also conducted collaborative testing on Llama 2, a large language model (LLM) developed by Meta. The tests were performed on [Amazon Elastic Compute Cloud \(Amazon EC2\)](#) G5 instances with NVIDIA A10G Tensor Core GPUs.

Another part of the development process involved making sure the LLM could recognize inappropriate questions from bad actors. Cyrus Chan, network specialist at HKBGE, says, "We need to safeguard the chatbot from misuse by individuals with disruptive intentions." This protection strategy was also part of a broader program to ensure the platform's resilience.

Working with AWS solutions architects, HKBGE built its trading platform on [AWS Well-Architected Framework](#) best practices, including using multiple [AWS Availability Zones](#) and [AWS Identity and Access Management \(IAM\) Identity Center](#) features for business continuity.

In addition, HKBGE is able to offer secure and reliable digital asset trading services to professional investors worldwide using AWS. "The AWS Availability Zones have provided high availability and assurance for our platform," says Chan. "We are confident that the AWS Well-Architected Framework has helped us build a robust and secure trading platform that can handle the high volume of transactions and inquiries we expect to see as we expand our business globally."



system architecture, which allows us to further develop the HKBGE trading platform without worries,” says Samuel Ng, tech operations director at HKBGE.

Today, financial services companies worldwide are adopting AWS solutions to accelerate innovation by delivering security and compliance at scale. In the case of HKBGE, it adopted [Amazon GuardDuty](#), [Amazon Inspector](#), [Amazon Detective](#), and [AWS Security Hub](#) to maximize cloud security. Wilson Ho, chief operating officer at HKBGE, says, “The AWS team gave us a lot of support around compliance. For instance, AWS compliance specialists discussed with us how to meet the recommendations of the Securities and Futures Commission in Hong Kong.”

The platform also employs services like [AWS Lambda](#), [Amazon Simple Notification Service](#) (Amazon SNS), AWS CloudWatch, and [AWS CloudTrail](#) for auditing. “Our integrated security approach on AWS provides real-time monitoring, ensuring adherence to regulatory standards and safeguarding the customer’s privacy,” adds Cyrus.

Outcome | Scaling Rapidly to Support Thousands of Expected Daily Inquiries

HKBGE is currently running its generative AI chatbot on AWS in a testing environment and plans to implement it with the launch of the trading platform, providing an excellent onboarding experience and platform for customers with personalized responses to FAQs. “Thanks to the flexibility of AWS, our generative AI chatbot can provide tailored answers to thousands of inquiries with speed and efficiency,” comments Wilson.

Once the chatbot is fully in production, HKBGE expects to reduce support costs for FAQs as much as 70 percent, freeing up resources to invest in staff training. Plus, the chatbot’s ability to operate 24/7 will provide round-the-clock personalized assistance to the company’s valued customers.

Beyond customer support, HKBGE plans to deploy the chatbot to address employees’ inquiries on internal processes and requirements. “We like the concept of the chatbot evolving into an internal knowledge base, providing snippets of information to enhance staff productivity,” says Wilson.

The HKBGE team is also exploring how generative AI can help customers in other potential aspects. “Generative AI on AWS has numerous potential use cases to enhance support for both our personnel and customers,” Wilson concludes.

About HKBGE

HKBGE is a financial technology company based in Hong Kong. It provides an efficient, fair and secure virtual asset trading platform for retail customers and professional investors who contribute to Hong Kong’s financial technology community.

AWS Services Used

Amazon EC2



Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

AWS Well-Architected

AWS Well-Architected helps cloud architects build secure, high-performing, resilient, and efficient infrastructure for a variety of applications and workloads.

[Learn more »](#)

Amazon GuardDuty

Amazon GuardDuty combines ML and integrated threat intelligence from AWS and leading third parties to help protect your AWS accounts, workloads, and data from threats.

[Learn more »](#)

AWS Security Hub

AWS Security Hub es un servicio de administración de la posición de seguridad en la nube (CSPM) que realiza revisiones de las prácticas recomendadas de seguridad, agrega alertas y permite la corrección automatizada.

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